UNIT 3 · TOPICS 3.14–3.15 · CHI-SQUARE TESTS

Chi-Square Tests for Homogeneity & Independence

Choosing the right test from the sampling design, writing hypotheses in context, verifying the three conditions, computing expected counts and the χ^2 statistic with $df = (r - 1)(c - 1)$, and stating a defensible conclusion.

1. Which Test? Homogeneity vs. Independence

Both tests use the **same** statistic, expected-count formula, conditions and degrees of freedom. They differ only in how the data were collected and what the claim is about.

	TEST FOR HOMOGENEITY	TEST FOR INDEPENDENCE
Design	Independent random samples from 2+ populations, or a randomized experiment with 2+ treatments	One random sample from a single population
Variables	One categorical response variable, compared across populations / treatments	Two categorical variables measured on each individual
Question	Is the distribution of the response the *same* across populations?	Are the two variables *associated* in the population?
H_0 in words	No difference in the distribution of [response] across [populations / treatments]	No association between [variable 1] and [variable 2] in [population]; the variables are independent
H_a in words	There is a difference in the distribution of [response] across [populations / treatments]	There is an association between [variable 1] and [variable 2] in [population]; they are not independent

DECIDE FROM THE SAMPLING DESIGN, NOT THE TABLE

Two identical-looking two-way tables can require different tests: if the row totals were **fixed by the researchers** (separate samples / treatment groups), it is homogeneity; if only the grand total was fixed and both variables were **observed on one sample**, it is independence. Always name the populations or variables in context.

2. Hypotheses, Conditions and the Chi-Square Distribution

THREE CONDITIONS (VERIFY ALL THREE, IN CONTEXT)

Randomization · 10% condition · all expected counts > 5

Randomization: independence needs a random sample; homogeneity needs independent random samples or a randomized experiment. **10% condition:** when sampling without replacement, $n \le 0.10N$ (unnecessary for a randomized experiment). **Expected counts:** every cell must have an expected count greater than 5 — this is about *expected*, not observed, counts.

The χ^2 statistic measures the distance between observed and expected counts relative to the expected counts. Chi-square distributions take only **positive** values and are **skewed right**; the skew becomes less pronounced as degrees of freedom increase. Large values of the statistic (far from zero) are evidence against H_0, so the p-value is always a **right-tail** area.

3. Expected Counts, the Statistic and Degrees of Freedom

THE THREE FORMULAS

$$E = (\text{row total})(\text{column total}) / (\text{table total})$$

$$\chi^2 = \sum (O - E)^2 / E \quad \text{over all cells}$$

$$df = (\text{number of rows} - 1)(\text{number of columns} - 1)$$

Expected counts are what we would anticipate *if H_0 were true*. Compute one for every cell (including the cells you did not observe), then sum the standardised squared discrepancies. The sum runs over all $r \times c$ cells; the degrees of freedom count the cells free to vary once the margins are fixed.

WORKED 2×2 EXAMPLE

Observed: [[42, 18], [30, 30]] with row totals 60, 60, column totals 72, 48 and table total 120. The expected count for (Tutoring, Passed) is $(60)(72)/120 = 36$; for (Tutoring, Failed) $(60)(48)/120 = 24$, and the second row has the same pair by symmetry. Then $\chi^2 = (42-36)^2/36 + (18-24)^2/24 + (30-36)^2/36 + (30-24)^2/24 = 1 + 1.5 + 1 + 1.5 = 5$ with $df = (2-1)(2-1) = 1$.

CALCULATOR COMMANDS (REAL TI-NSPIRE NAMES)

- `chi22way obsMatrix` — the test for association from a two-way observed-count matrix.
- `χ2Cdf(5,9E999,1)` — right-tail p-value for $\chi^2 = 5$, $df = 1$.
- `invχ2(0.95,2)` — the 5% critical value with 2 degrees of freedom.

4. P-Value, Conclusion Template and Exam Traps

P-VALUE AND DECISION

$$\text{p-value} = P(\chi^2_{df} \geq \chi^2_{observed})$$

Compare with α. In the worked example $P(\chi^2_1 \geq 5) \approx 0.025$, so at $\alpha = 0.05$ we reject H_0.

EXAM-SAFE CONCLUSION

"Because the p-value of [value] is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence of [an association between the two variables in the population **/** a difference in the distribution of the response across the populations]." If the p-value is large, say "we fail to reject H_0; we do not have convincing evidence..." — never "accept H_0" or "prove independence".

TRAPS THE EXAM ACTUALLY TESTS

- **Wrong test name / wrong context:** homogeneity names populations or treatments; independence names the two variables and one population.
- **Observed vs. expected counts:** the > 5 condition and the E formula use expected counts; observed counts go in O.
- **Use counts, not proportions** in the statistic — percentages make χ^2 wrong.
- **df is not $n - 1$** and not the number of cells; it is $(r - 1)(c - 1)$.
- **The p-value is always the right tail**; a large χ^2 is evidence against H_0.
- **Association is not causation** for an observational sample; a chi-square test never establishes which variable causes the other.
- **A significant result does not say which cell differs** — compare the observed and expected counts to describe the largest contributions.